PROMPT ENGINEERING

LAB EXPERIMENT

To evaluate and compare the effectiveness of prompting techniques (zero-shot, few-shot, chain-of-thought, role-based) across different AI platforms (e.g., ChatGPT, Gemini, Claude, Copilot) in a specific task: text summarization.

SLOT NO:T1-S5

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Blockchain Technology

Blockchain technology is a modern digital system used to store and record information in a secure, transparent, and decentralized way. Instead of storing data in a single central database, blockchain distributes the data across a network of computers called **nodes**. Each participant in the network has a copy of the blockchain, which makes the system highly secure and difficult to manipulate.

A **blockchain** is made up of a sequence of blocks that are linked together in chronological order, forming a chain. Each block contains three main components: **data**, a **hash**, and the **hash of the previous block**. The data stored in a block can include information such as transaction details, timestamps, or records. The hash is a unique digital fingerprint that identifies each block. The hash of the previous block connects all the blocks together, creating a continuous and secure chain.

One of the most important features of blockchain technology is **decentralization**. In traditional systems, data is controlled by a central authority such as a bank or a company. In contrast, blockchain removes the need for a central authority because all participants in the network collectively verify and maintain the records. This increases transparency and trust among users.

Another key feature is **immutability**. Once data is recorded in a blockchain block and confirmed by the network, it becomes extremely difficult to change or delete. If someone attempts to alter the data in one block, the hash of that block will change, breaking the chain and making the tampering easily detectable. This makes blockchain highly secure and reliable for storing sensitive information.

Blockchain also uses **consensus mechanisms** to validate transactions. A consensus mechanism is a method used by the network participants to agree on the validity of transactions before they are added to the blockchain. Common consensus methods include **Proof of Work (PoW)** and **Proof of Stake (PoS)**. These mechanisms ensure that all transactions are verified and that the network remains trustworthy.

The most well-known application of blockchain is in **cryptocurrencies**, such as Bitcoin. However, blockchain technology has many other applications beyond digital currency. It can be used in areas like **supply chain management**, **healthcare**, **digital identity verification**, **voting systems**, and **smart contracts**. Smart contracts are self-executing agreements written in code that automatically carry out actions when certain conditions are met.

In conclusion, blockchain technology is a powerful innovation that provides secure, transparent, and decentralized data management. Its ability to ensure trust without relying on a central authority makes it useful in many industries. As technology continues to evolve, blockchain is expected to play a significant role in transforming digital transactions and information management in the future.

FEW SHOT PROMPT IN CHATGPT:

Prompt:

"Summarize the following passage about blockchain technology in 4–5 sentences highlighting the key points."

RESPONSE:

ZERO-SHOT PROMPT IN CHATGPT:

Prompt:

"Summarize the following passage about blockchain technology in 4–5 sentences highlighting the key points."

RESPOSE:

Blockchain technology is a system used to store and record information securely in a decentralized network of computers. Data is stored in blocks that are connected using cryptographic hashes to form a secure chain. Key features of blockchain include decentralization, transparency, and immutability, which make the data difficult to

change or manipulate. Transactions are verified through consensus mechanisms such as Proof of Work and Proof of Stake. Blockchain is widely used in cryptocurrencies like Bitcoin and also has applications in supply chain management, healthcare, digital identity, voting systems, and smart contracts.

CHAIN OF THOUGHT IN CHATGPT:

Prompt:

"Read the following passage about blockchain technology. Think step-by-step to identify the main idea, key features, and applications. Then write a clear summary of the passage in 4–5 sentences highlighting the important points."

> **Response (Chain-of-Thought Style Summary):**
>
> **First, the passage explains that blockchain technology is a digital system used to store and record information securely in a**

decentralized network of computers. Next, it describes how blockchain works by linking blocks that contain data, a hash, and the previous block's hash, forming a secure chain of records. The passage then highlights key features such as decentralization, transparency, and immutability, which make the system trustworthy and difficult to manipulate. It also explains that consensus mechanisms like Proof of Work and Proof of Stake are used to verify transactions. Finally, the passage concludes that blockchain is widely used in cryptocurrencies like Bitcoin and has applications in areas such as supply chain management, healthcare, digital identity, voting systems, and smart contracts.

ROLE-BASED PROMPT IN CHATGPT:

Prompt:

"You are a technology teacher. Read the following passage about blockchain technology and summarize it in 4–5 clear

sentences highlighting the definition, key features, and applications."

Response:

As a **technology teacher**, I would explain that **blockchain technology** is a digital system used to store and record data securely in a decentralized network of computers. Information is stored in blocks that contain data, a unique hash, and the hash of the previous block, which links the blocks together to form a secure chain. Important features of blockchain include **decentralization, transparency, and immutability**, making the data difficult to alter or manipulate. Transactions are verified using **consensus mechanisms such as Proof of Work (PoW) and Proof of Stake (PoS)**. While blockchain is widely known for supporting **cryptocurrencies like Bitcoin**, it is also used in areas such as supply chain management, healthcare, digital identity verification, voting systems, and smart contracts.

PROMPT IN DEEP SEEK FOR FEW-SHORT:

"Provide a brief summary of the main features and uses of blockchain mentioned in this article using bullet points."

RESPONSE:

Blockchain is a secure, decentralized digital ledger distributed across a network of computers. It records data in linked blocks, ensuring transparency and immutability. By removing the need for a central authority and using consensus mechanisms for verification, it provides a trustworthy system for applications like cryptocurrencies, supply chains, and secure data management.

PROMPT IN DEEP SEEK FOR ZERO-SHORT:

Prompt:

"Summarize the following passage on Blockchain Technology in two sentences."

RESPONSE:

Blockchain is a decentralized digital ledger that distributes data across a network of computers, eliminating the need for a central authority and making the system highly secure and transparent. It structures information into interconnected blocks using unique hashes, which ensures that once data is recorded, it becomes immutable and tamper-proof.

PROMPT IN DEEP SEEK FOR CHAIN OF THOUGHT:

Prompt:

"I want you to summarize the following passage on Blockchain Technology. First, identify the core definition. Second, list the three key features mentioned. Third, note the major applications. Finally, combine these insights into a single, cohesive summary paragraph."

RESPONSE:

Step 1: Identify the Core Definition

Blockchain technology is a modern digital system for storing and recording information in a secure, transparent, and decentralized manner. Instead of relying on a single central database, it distributes data across a network of computers (nodes), where each participant holds a copy of the ledger.

Step 2: List the Three Key Features

1. **Decentralization:** It removes the need for a central authority (like a bank) by allowing all network participants to collectively verify and maintain records, which increases trust.
2. **Immutability:** Once data is recorded and confirmed in a block, it becomes extremely difficult to change or delete. Any tampering attempt alters the block's unique hash and breaks the chain, making fraud easily detectable.
3. **Consensus Mechanisms:** The network uses methods like Proof of Work (PoW) and Proof of Stake (PoS) to validate transactions, ensuring that all participants agree on the

validity of the data before it is added to the ledger.

Step 3: Note the Major Applications
While the most famous application is cryptocurrencies (like Bitcoin), the technology is also used in supply chain management, healthcare, digital identity verification, voting systems, and smart contracts (self-executing agreements written in code).

Step 4: Combine Insights into a Cohesive Summary
Blockchain is a decentralized digital ledger that distributes data across a network of computers, ensuring security and transparency without relying on a central authority. Its structure of linked, hash-protected blocks guarantees immutability, while consensus mechanisms like Proof of Work validate transactions, making it a reliable foundation not only for cryptocurrencies like Bitcoin but also for applications in supply chain management, healthcare, and smart contracts.

PROMPT IN DEEP SEEK FOR ROLE-BASED:

Prompt:

"Act as a high school computer science teacher. Summarize the blockchain passage below in a way that is easy for students to understand. Use simple explanations and highlight the key terms they need to remember for an exam."

Response (as a High School Computer Science Teacher):

Alright, class, settle down. Let's talk about **Blockchain**. I know you've heard the word thrown around with Bitcoin, but it's actually a really cool way to store information. Think of it as a high-tech digital notebook that everyone in the class can see, but no one can erase.

Here is the simple breakdown you need to remember for the exam:

1. What is it? (The Basics)

Instead of storing data in one secret spot (like a school principal's locked filing

cabinet), a blockchain spreads copies of the data across thousands of computers all over the world. We call these computers **Nodes**. Because everyone has a copy, it's super hard for a hacker to cheat the system—they'd have to break into every single computer at the exact same time.

2. How is it built? (The Chain)
Imagine a LEGO tower.

- **Blocks:** Each LEGO brick is a **Block**. It holds data (like who paid whom and when).
- **Hashes:** Every brick has a unique fingerprint called a **Hash**. If you try to change the data inside the brick, the fingerprint changes completely.
- **The Glue:** Each new brick contains the fingerprint (hash) of the brick before it. This is what links them together into a **Chain**. If a hacker tries to mess with an old brick, its fingerprint changes, and the whole chain breaks—and everyone notices immediately!

3. Why is it special? (Key Features to Memorize)

- **Decentralization:** Normally, a bank or a company is the boss of your money. With blockchain, there is no single boss. The group (the nodes) verifies everything together. It's like a group project where everyone has to agree on the final grade.
- **Immutability:** This is a big vocabulary word. It means **unchangeable**. Once something is written into a block and added to the chain, you cannot delete it or change it. It's there forever.

4. How do we trust it? (Consensus)
The computers on the network have to agree before adding new info. They use math problems to verify transactions. The book mentions two ways: **Proof of Work (PoW)** and **Proof of Stake (PoS)** . Just know that these are the rules the computers follow to agree on what is true.

5. What is it used for?
Yes, it runs **Cryptocurrencies** like Bitcoin. But it's also used to track organic food in the grocery store, secure medical records, and run **Smart Contracts**—which are basically

automatic agreements that execute themselves when conditions are met (like a vending machine giving you a soda when you put money in).

Exam Tip: Remember the three pillars: **Decentralization, Immutability, and Transparency.** Blockchain creates trust between people who don't know each other by using code and math instead of a middleman.

PROMPT IN GEMINI FOR FEW-SHORT PROMPT:

PROMPT:

"Summarize this passage in three bullet points focusing on how the technology works and why it's secure."

RESPONSE:

Based on the passage provided, here is the summary of blockchain technology focused on its mechanics and security:

- **Decentralized Data Structure:** Instead of a central server, data is distributed across a global network of **nodes**; each participant maintains a full copy of the ledger, ensuring no single entity has total control.
- **The Cryptographic Chain:** Information is stored in blocks linked by **hashes** (unique digital fingerprints); because each block contains the hash of the previous one, any attempt to alter past data breaks the entire chain's sequence, making tampering immediately obvious.
- **Validation via Consensus:** New transactions are only added once the network agrees on their validity through **consensus mechanisms** (like Proof of Work), which removes the need for a middleman and prevents fraudulent entries.

PROMPT IN GEMINI FOR ZERO-SHORT PROMPT:

"Summarize the provided text on blockchain technology into a concise paragraph. Focus on the core definition, the three main components of a block, and the significance of decentralization and immutability."

RESPOSE:

Blockchain technology is a decentralized digital ledger that records data across a network of nodes, eliminating the need for a central authority and enhancing transparency. Each block in the chain consists of three essential elements: the **data** (such as transaction details), a **unique hash** acting as a digital fingerprint, and the **hash of the previous block**, which cryptographically links the sequence together. The significance of this architecture lies in its **decentralization**, which fosters collective trust among participants, and its **immutability**, which ensures that once information is validated via consensus mechanisms like Proof of

Work, it becomes nearly impossible to alter without breaking the entire chain.

PROMPT IN GEMINI FOR CHAIN OF THOUGHT :

PROMPT:

“Summarise the prompt based on chain of thought from the above passage”

RESPONSE:

tep 1: Identify Core Technical Pillars

First, I need to extract the fundamental concepts that make the system function:

- **Decentralization:** Data is stored across multiple nodes (computers) rather than a central server.
- **Structure:** A linear chain of blocks containing data, a unique **Hash**, and the **Previous Hash**.
- **Immutability:** The "digital fingerprint" (hash) ensures that if one block is changed, the entire chain breaks.

- **Consensus:** Mechanisms like Proof of Work (PoW) ensure everyone agrees on what is true.
- **Utility:** Applications range from Bitcoin to self-executing Smart Contracts.

Step 2: Analyze Logical Connectivity

Next, I'll look at how these pieces interact to create security: The **Previous Hash** is the "glue." Because Block B contains the hash of Block A, they are mathematically linked. If a bad actor changes Block A, its hash changes. Since Block B no longer recognizes that new hash, the connection fails. This creates a "trustless" environment where you don't need a bank; you only need the math to be correct.

Step 3: Synthesis of Complex Ideas

Blockchain is a distributed ledger technology that replaces central authorities with a network of peer-to-peer nodes. It relies on a cryptographic "chain" where each block is anchored to the one before it. By using consensus protocols to validate data, it

ensures that once information is written, it is permanent (immutable) and transparent to all participants.

Step 4: Final Summary (The "Layman" Version)

Blockchain is a secure, decentralized digital record-keeping system that stores information across a network of computers instead of a single central bank or company. Each piece of data is locked into a chain using unique digital fingerprints, making it nearly impossible to hack or change without being caught. This technology allows for transparent transactions in everything from cryptocurrency to healthcare and legal contracts.

PROMPT IN GEMINI FOR ROLE-BASED:

PROMPT:

"Act as an expert Technology Educator. Your task is to summarize the provided passage on Blockchain Technology for a class of

undergraduate students. Please follow these constraints:

1. The Anatomy of a Block

At its core, a blockchain is a chronological string of data packages. For a block to exist, it must contain three specific elements:

- **The Data:** The actual information being recorded (e.g., a digital contract or a financial transaction).
- **The Hash:** Think of this as a unique digital fingerprint. If even a single character of the data changes, the hash changes completely.
- **The Previous Hash:** This is the "hook" that connects the current block to the one before it.

2. Security through Architecture

The system achieves high-level security through two primary pillars:

- **Decentralization:** Unlike a traditional bank that keeps one master ledger, blockchain uses a **Peer-to-Peer (P2P) network**. Every participant (node) has a full copy. To "hack"

the system, you would have to compromise the majority of the network simultaneously, which is computationally impractical.

- **Immutability:** Because each block references the **Previous Hash**, the chain is mathematically linked. If a malicious actor alters Block A, its hash changes. This causes a mismatch in Block B, effectively "breaking" the chain and alerting the entire network to the tampering.

3. Validation via Consensus

To ensure the ledger remains honest without a central boss, the network uses **Consensus Mechanisms**.

- **Proof of Work (PoW)** and **Proof of Stake (PoS)** are the protocols that nodes use to agree that a transaction is valid before it is officially added to the chain. This collective verification creates "trustless trust"—you don't need to know the other person; you just need to trust the protocol.

4. Beyond Cryptocurrency

While Bitcoin made blockchain famous, the utility of the technology is expanding. We are seeing a shift toward:

- **Smart Contracts:** Self-executing code that triggers automatically when conditions are met (e.g., releasing a payment only when a shipment is scanned at a warehouse).
- **Industry Integration:** Applications in healthcare records, voting integrity, and transparent supply chain tracking.

EVALUATION TABLE

Prompt technique	Platform	Accuracy	Coherence	Simplicity	Speed	UX
Few -shot	CHATGPT	5	5	3	4	4
Zero-shot		4	4.5	4	3.5	4
Chain of thought		4	4	3	5	4
Role-based		4.5	3	4	3	3
Few -shot	DEEP SEEK	5	3.5	4	5	5
Zero-shot		4	5	4	4	3.5
Chain of thought		3	3	3.5	4	3
Role-based		4	4.5	5	4	4
Few -shot	Gemini	5	5	4	4	4

Zero-shot		4	3.5	4	3	5
Chain of thought		5	4	3	5	4
Role-based		4	5	3.5	4	3